PART 1: DISCOVERY, PROBLEM SPACE

What is a VMS?

A Vendor Management System is software that helps a company handle its vendors from start to finish: signing on a new supplier, tracking how they perform, renewing contracts, and eventually ending the relationship. The whole idea is to keep everything about a vendor in one place, instead of spreading across different tools, spreadsheets and email.

That is what separates it from a smaller, single-job tool. A contract tool just handles contracts. A payment tool just handles payments. A VMS is meant to sit above all of them and give one clear view of the full vendor journey. One quick note on the word: "VMS" gets used two ways. Sometimes it means this broad vendor software, and sometimes, mostly in staffing, it means a tool just for managing contractors. This case study is about the broad meaning.

Why this space, and why now

Companies today rely on a lot of outside vendors, sometimes hundreds or thousands, for software, legal help, shipping, marketing and hiring. They have bought plenty of tools to manage the different parts. But when a new vendor needs to be set up, all those tools still do not add up to one smooth process. The work falls back into emails and spreadsheets, and nobody can see where things stand. That gap, lots of tools but no single visible workflow, is what makes this a real opportunity.

The market backs up why leadership sees this as worth chasing. The vendor management software market was worth about USD 11.6 billion in 2025 and is expected to reach USD 30.9 billion by 2034, driven by companies going more digital and wanting to automate onboarding. The neighbouring procurement software market is growing the same way, from about USD 8.9 billion in 2025 toward USD 20.8 billion by 2034. It is a big, fast-growing space, and onboarding sits right at the heart of the pain.

1.1 Domain immersion: mapping the space

We started wide, reading job postings, real user reviews on G2 and Capterra, practitioner threads, and India's compliance material, before forming any opinion. Two things came out of it: a working vocabulary, and a map of where in the vendor lifecycle the noise concentrates.

Glossary (selected terms)

Term	In plain English	Why it matters
Vendor onboarding	Setting up a new supplier in your systems so you can legally transact and pay them	The step everything else depends on; our focus
Vendor master data	The saved record of a vendor (tax, bank, contact, compliance details) in the ERP	If this is wrong or duplicated, invoices fail and payments stall

MSA / SOW / NDA	Master agreement, statement of work, and non-disclosure; the core contract chain	The contracting stage our interviews named as the biggest bottleneck
GRN (goods receipt note)	Proof that goods or services were received	Without it, finance cannot safely pay; a manual gap in most SMEs
Three-way match	Checking PO, GRN and invoice agree before paying	Where finance catches errors; breaks when data is scattered
KYC / TIN / sanctions check	Identity, tax-ID and watchlist verification of a vendor	The compliance gate before a vendor is allowed to transact
MSME 45-day rule (Sec 43B(h))	Indian tax rule disallowing a deduction if an MSME vendor is paid after 45 days	Makes verifying MSME status at onboarding a financial, not optional, step
Source-to-pay (S2P)	The full chain from finding a vendor to paying them	The scope the big enterprise suites sell; onboarding is only one slice

Map of the space

Breaking the theme into sub-areas shows where the complaints cluster. Onboarding and the visibility around it draws far more noise than any other stage.

Sub-area	What happens here	What seems to break	Noise
Sourcing & discovery	Finding and shortlisting vendors	Informal, relationship-driven; no record	Low
Onboarding & compliance	Collecting docs, verifying, approving	Scattered, slow, no visibility into status	High
Contracting	NDA, MSA, SOW negotiation and signing	Legal-to-legal back-and-forth stalls everything	High
Purchase orders	Raising and approving POs	Assumes vendor already onboarded	Medium
Delivery & GRN	Confirming goods/services received	Manual, trust-based, often skipped	Medium
Invoicing & payment	Matching and paying invoices	Double entry between GRN and AP	Medium
Performance & renewals	Tracking SLAs, renewing	Runs on spreadsheets, disconnected	Medium

1.2 The current ecosystem: lots of tools, no single view

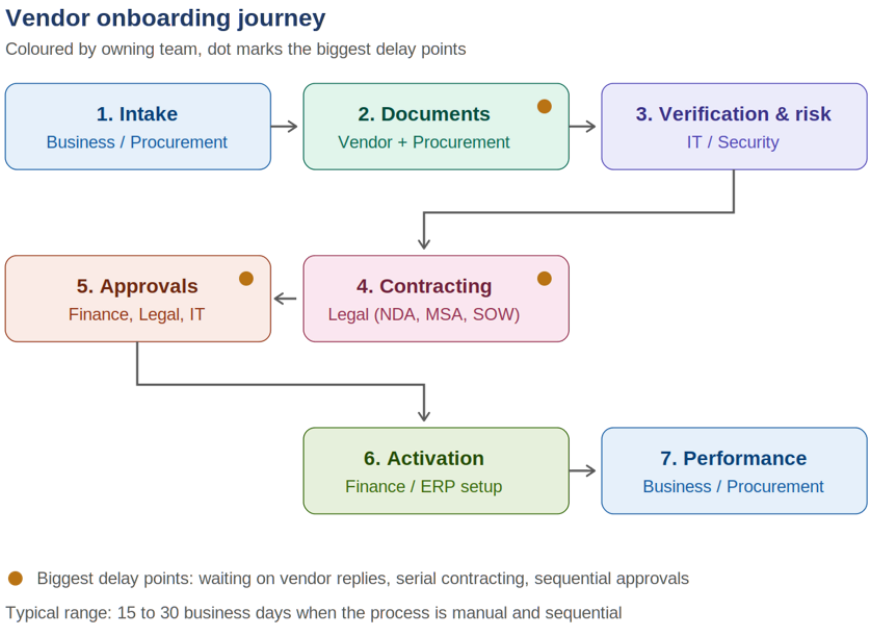
Onboarding rarely breaks because a tool is missing. It breaks because it is the seam between six or seven tools, each bought to solve a different team's problem. Laid out by stage, the gaps are easy to see:

Stage	Who owns it	Typical tools	The gap
Sourcing	Business / Procurement	LinkedIn, referrals, informal outreach	No record until a vendor is picked
Onboarding & intake	Procurement; Legal & IT gate it	Email + Excel/Forms; portals inside Ariba/Coupa at the top end	No shared intake all teams trust; data re-keyed 3-4 times
Contracting	Legal	Word redlines, DocuSign; Icertis/Ironclad at the top end	NDA, MSA, SOW each crawl through their own review

Purchase orders	Procurement	SAP Ariba, Coupa, Oracle, Zycus, Ivalua	These assume onboarding is already done
Payments	Finance / AP	SAP, NetSuite, Tipalti, Zoho Books	Bank/tax checks duplicate what Legal already collected
Risk & compliance	IT / Security / Legal	OneTrust, Prevalent (separate systems)	No shared risk tier other workflows can see
Performance	Business / Procurement	Spreadsheets, mostly	Nothing links back to the onboarding record

Two patterns stood out. ***First***, even where an enterprise tool exists, the real work still runs on spreadsheets and email, the job postings we read say "proficient in vendor management software" and then list Excel in the same breath. ***Second***, the big suites that hold most of the market (SAP Ariba, Coupa, Oracle, GEP, Ivalua, Zycus, together roughly 59% of procurement software) are built for large enterprises that can absorb a 6-to-18-month rollout and a dedicated procurement team. Real users describe them bluntly: on r/procurement, Coupa gets called overpriced and overly complex, and one comment about Ariba said it was not designed to be user friendly or intuitive at all. That complexity is the opening: mid-sized teams cannot realistically adopt these, so they fall back to email and lose all visibility.

Mapping the journey visually makes the scattering obvious. A single vendor passes through at least seven stages, each owned by a different team, and the work snakes back and forth across the organisation with no shared view of where it has reached:



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This is the black box in one picture: seven owners, three delay points, and no single place anyone can look to see the status. The delay range on the diagram is a symptom; the real problem is that the work is invisible while it moves.

1.3 Narrow To Three Candidate Problems

Hypothesis 1: *Vendors & buyers struggle to gain visibility & traceability because both the vendor and the buyer don't know the exact status of the vendor onboarding process, which leads to delays, manual dependencies, and lack of transparency. Onboarding and information management is the largest revenue segment (about 30%) within VMS software; analytics and reporting is the fastest growing (roughly 11.5% CAGR). 73% of procurement leaders struggle with manual data, yet most still rely on it for onboarding*

Hypothesis 2: *Vendors & buyers struggle to retrieve, follow up and maintain contracts and other documents because they primarily use emails, which leads to considerable manual efforts and increases chances of manual errors, incorrect documentation, missed timelines, reminders and delays in retrievals. 71% of Companies Cannot Locate at Least 10% of Their Active Contracts.*

Hypothesis 3*: Buyers struggle with verifying the vendor and their related documentation because they have to manually verify and validate the documents, which not only increases the manual time and effort required, but also leads to risks associated with regulatory and government compliance due to insufficient validations and incorrect documentation. Research indicates 53% of procurement professionals rate their supplier data quality as poor, with none rating it excellent.*

Amongst the above identified hypothesis and based on the initial research and criticality in the vendor management lifecycle, we believe improving the visibility and traceability will have a bigger impact and hence we directed our future research towards this.

1.4 Secondary Research On Your Chosen Problem

Even though a Vendor management system (VMS) has multiple workflows, onboarding is one of the crucial processes as this is the step which acts as an input to next steps. Many applications available in the market address the issues related to vendor onboarding but the issues are not completely resolved though.

#	Finding	Why this matters to us	Source
1	The systems do not talk to each other and data is spread across	The vendor onboarding data doesn't flow across the systems. Many times, the data is stored in old ERPs, excelsheets.	Coherent Market Insights
2	Tools are never built as one source of truth	Teams spend a lot of time jumping between tools and the tools are not designed to hold all data at one place. So once there are more than 20-30 vendors, it becomes difficult to manage.	Unity Connect IntelmarketresearchIntel marketresearch

3	Upon onboarding the vendor, visibility is a challenge	Tools are good at getting vendors 'in the door' but after that tracking the processes like approval, document submission etc is cumbersome.	Gatekeeper Mindsprint
4	Different teams are looking for different information	Procurement cares about price, legal cares about risk, business teams just want speed. When nobody's aligned, people start skipping the process entirely, like a team just signing up for a tool on their own because waiting for approval takes too long.	Technology Match
5	Good VMS tools are expensive, so small and mid-sized companies get left out	The big platforms (SAP Fieldglass, Coupa, Ivalua, etc.) are built for large enterprises with big budgets and big teams. Smaller companies often can't afford the cost or the effort needed to set them up, which honestly leaves an opening for a leaner, simpler tool.	Intel Market Research

The outcome of secondary research is that the vendor onboarding process is broken with dispersed and disconnected data, lower visibility and lack of ownership.

1.5 Primary Research

We interviewed seven people across the vendor lifecycle: a Site Admin Manager, an SME Manufacturer in Finance and Operations, an HR Lead, a Sales Head, a Factory Owner (bags manufacturing), an Engineering Lead at a drone startup, and a Senior Finance Executive working on SAP. The table below merges the context of each conversation with what we took away from it.

#	Persona	Context	Key findings
1	Site Admin Manager	Took us through the full process (discovery, vetting, agreement, PO, onboarding), which takes 15 days to 3 months	Finding vendors is manual and based on relationships. The legal-to-legal stage is the slowest part. Document checks rarely block things, but sometimes do
2	SME Manufacturer (Finance/Ops)	Uses Zoho Books only for payments; everything else (SLAs, negotiations, MSAs) is offline over email	Vendor base (MSMEs, hospitals) is too non-tech-savvy for automation; budget blocks it too; no visibility between MSA signing and delivery, status is trust-based
3	HR Lead	Sources LMS and HR vendors through social groups and demos, with light documentation (SLA, NDA, ISO 27001)	Sourcing relies on informal channels; documentation is minimal and security-focused; adoption and compliance after onboarding are harder than onboarding itself
4	Sales Head	Described vendor registration from the seller side, plus plans for CRM-Zoho integration	Registration docs (GST, PAN, TAN, MSME, bank) are fairly standardized; the buyer's decision level stays opaque until late; manual today, moving toward CRM
5	Factory Owner (bags)	Onboards around 5 vendors a month	Represents high-volume onboarding, a contrast to the lower-frequency SME cases
6	Engineering Lead (drone startup)	NDA-limited discussion	Confirms confidentiality-heavy onboarding in a defence-sector supply chain

7	Finance Senior Executive (SAP)	Discussed SAP migration, cost model and workflow gaps	Manual double-entry between GRN and AP; SAP's tcode-based licensing limits team access; no systematic way to track goods received or tell vendors to stop excess shipments

The same thing kept coming up across very different businesses. The real complaint wasn't speed, it was not being able to see where a vendor had reached or what was holding it up. The SME Manufacturer said status was based on trust, with no view between signing and delivery, and even the team already on SAP had no clean way to track what had arrived. So the interviews point to the same problem the secondary research did, just from the inside: onboarding is scattered, status is hidden, and the tools meant to fix it are either too basic or too complex to be useful.

1.6 Synthesis

#	Pain Point	Source	Severity	Evidence, direct quote or observation
1	No Single source of truth	Primary: SME Manufacturer, Finance Exec (SAP); secondary research	High	Vendor data sits in old ERPs and Excel sheets, and once there are more than 30 vendors it gets hard to manage. Zoho Books is used only for payments, and the SAP team still keys data in twice between GRN and AP, so no single system holds the full picture
2	Lack of visibility/status tracking during and after onboarding	Primary: SME Manufacturer, Finance Exec (SAP); Hypothesis 1; secondary research	High	Neither the vendor nor the buyer knows exactly where an onboarding stands. The SME Manufacturer said status was trust-based with no view between MSA signing and delivery, and the SAP team had no way to track what was received. Tools get vendors "in the door" but tracking after that is hard
3	Manual, email-dependent contracting/negotiation and document handling	Primary: SME Manufacturer, HR Lead; Hypothesis 2	High	SLAs, negotiations and MSAs are all handled over email, and paperwork is minimal. Leaning on email means a lot of manual work and plenty of room for mistakes
4	Manual vendor/document verification & compliance	Primary: Site Admin Manager, HR Lead; Hypothesis 3	Medium	Document checks rarely block things but sometimes do. Getting people to use the vendor and stay compliant after onboarding is harder than onboarding itself, and there are legal and government compliance risks
5	Cross Team misalignment	Primary: Site Admin Manager; secondary research	Medium	Procurement cares about price, Legal about risk, and business teams just want speed. The legal-to-legal stage was named the slowest part. When no one is aligned, people start skipping the process altogether

What surprised us

We expected the main complaint to be speed, that onboarding just takes too long. What surprised us was that the bigger frustration was visibility. People couldn't see where a vendor was stuck, and the delay turned out to be a symptom of that, not the real problem.

What we believed going in that turned out to be wrong

We thought companies avoided the big tools like SAP Ariba and Coupa mainly because of the price. The SAP finance executive showed us the real issue: even when a company owns one, it's too complex for people to actually use, so work slips back to email. The barrier is complexity, not just cost.

Who does not have this problem

Two groups don't really feel this pain. Big enterprises with dedicated procurement teams already run a tool like SAP Ariba and have the people to manage it. And very small or owner-run businesses that onboard just a few familiar vendors, like the factory owner's high-trust relationships, do fine on phone and WhatsApp, because the volume is low and the trust is high. The pain is worst for the mid-sized company in between: too many vendors for email and trust to keep up, but too small and not technical enough to take on an enterprise tool.

1.7 Problem Statement

A Procurement Ops Manager at a company onboarding more than 30 vendors a month struggles to get each vendor live quickly, because approvals across HR, Legal, Finance and Procurement are coordinated over email and spreadsheets with no single view of status, and the existing tools are too complex for non-technical users to adopt. This costs them 15-30 business days per vendor, plus constant manual chasing and no accountability for where a request is stuck.

Why now

Many startups and mid-sized enterprises have scaled to onboarding a large number of vendors, and they own a separate tool for each slice of the lifecycle, yet none stitches onboarding into one shared, visible workflow. That gap, many tools but no single process, has only recently grown big enough to be worth solving.

PART 2: DISCOVERY, SOLUTION SPACE

2.1 How This Is Solved Today

Every problem already has a solution, even if it is a spreadsheet or a WhatsApp group. Below is what companies actually use today, including the non-product workarounds, which could be our real competition.

Current solution	Who uses it	What works	What breaks	What it costs
Email + spreadsheets + shared drives	SMEs and most mid-sized teams	Zero setup, everyone already knows it, infinitely flexible	No shared status, data re-keyed across teams, nobody can see where a vendor is stuck, the black box	"Free," but 15-30 days per vendor and constant manual chasing
Point tools (Zoho Books, DocuSign, Google Forms)	SMEs stitching a stack together	Each does its one job well (payments, signing, intake)	Tools don't talk to each other, no single source of truth, status still lives in inboxes	Low per-tool, but integration and reconciliation effort is high
Enterprise S2P suites (SAP Ariba, Coupa, Ivalua, Fieldglass)	Large enterprises with procurement teams	Genuinely end-to-end, strong controls once configured	Too complex and costly; 6-18 month rollouts; non-technical users can't adopt them, so work leaks back to email	High licence + implementation cost; out of reach for SMEs
"They just live with it" (phone + WhatsApp chasing)	Owner-run SMEs, high-trust vendor relationships	Fast for a handful of known vendors; human judgement	Status is trust-based, breaks past ~30 vendors, no audit trail, no compliance record	Hidden cost in owner/manager time and compliance risk

The pattern: the cheap options give up all visibility, and the option that gives visibility (enterprise suites) is too complex and expensive for the teams that feel the pain most. That gap is exactly what our solution targets.

2.2 What You Chose And Why

- **Chosen approach:** A role based vendor onboarding system that allows different stakeholders to easily collect, and verify the information and take necessary action. The workflow has complete visibility across all stakeholders.
- **Why this one:** ability to configure user access, based on roles, provides flexibility to decide who will contribute to the vendor onboarding process and how. The workflow allows inviting potential vendors to upload required documents and information, which flows across the departments. This build visibility across different departments and users, also reduces time taken to complete the onboarding process. Eventually the system becomes a single source of truth for vendor onboarding flow.

What's not covered in v1, and why:

- Vendor self-registration. Onboarding is invite-only (buyer-initiated). Letting any vendor self-register invites spam and duplicate records, and our interviews showed onboarding is always sponsored by an internal team.
- The full vendor lifecycle (RFP, PO, delivery, invoicing, performance, offboarding). We are solving onboarding only, because it is the input every later step depends on and it is where the pain concentrated. Trying to solve the whole lifecycle is what makes the enterprise suites unadoptable.

- Real AI document review. Shown only as a mocked, toggle-able suggestion chip. The research signal for AI was low-confidence, and building real OCR/risk-scoring would blow the scope without addressing the core visibility problem.
- Payroll, native mobile apps, multi-language, and IT/support ticketing. Each is a separate product surface that adds complexity without moving the visibility metric. Mobile is covered by responsive web instead.

PART 3: THE SPECIFICATION

The MVP is a self-service Vendor Onboarding Portal that automates the collection, validation and approval of vendor information (business details, tax documentation, banking) with role-based access and full status visibility.

3.1 Goals, Success Metrics

Primary metric: the one number that tells you this worked.

Metric	Definition	How it is measured	Baseline	Target
Onboarding cycle time	Median business days from vendor invite accepted to final Admin approval	Timestamp on invite-accept event and on final-approval event, both stored on the vendor record; difference reported	15-30 days (manual, per case brief)	Under 7 business days for a standard vendor

Why this metric: it is the direct expression of the problem (slow, invisible onboarding) and it is measurable from two events we already fire. It is not a vanity metric like signups.

Supporting product metrics (from the prototype analytics):

- Number of onboardings completed per period, filterable by requesting team.
- Department-wise onboarding speed versus that department's SLA.
- Number of SLA breaches per department (excluding paused-clock time during resubmissions).
- Average time the vendor takes to submit all mandatory documents, and to resubmit after a clarification.
- Acceptance versus rejection rate of onboardings.

Guardrail metrics: things that must not get worse while we chase speed.

Guardrail	Threshold that would worry us
Rejected/halted vendors slipping through	Any vendor onboarded while a department flag is still open
Compliance completeness	Any vendor activated with a mandatory document missing or expired
Review quality	Approval time per department dropping so low it implies rubber-stamping

3.2 Scope

In scope for v1:

- ***Role-based access control (RBAC)***, vertical (by role) and horizontal (by department), enforced at route and data level.
- ***Web portal***, desktop and mobile-responsive, for internal users and vendors.
- ***Admin***: configure onboarding, invite vendors, set SLAs and notification triggers, manage team access, halt onboarding, give final approval, view analytics.
- ***Buyer departments*** (HR, Legal, Finance, Procurement): view assigned vendor requests read-only, approve or reject with mandatory comment on reject, flag a vendor to Admin, see the end-to-end status bar.
- ***Vendor***: sign up via invite + OTP, complete the onboarding form, upload documents, submit, see real-time status (department-wise and document-wise), resubmit with clarification.
- ***SLA logic*** including pause-on-resubmission, and in-app notifications on key triggers.

Out of scope for v1:

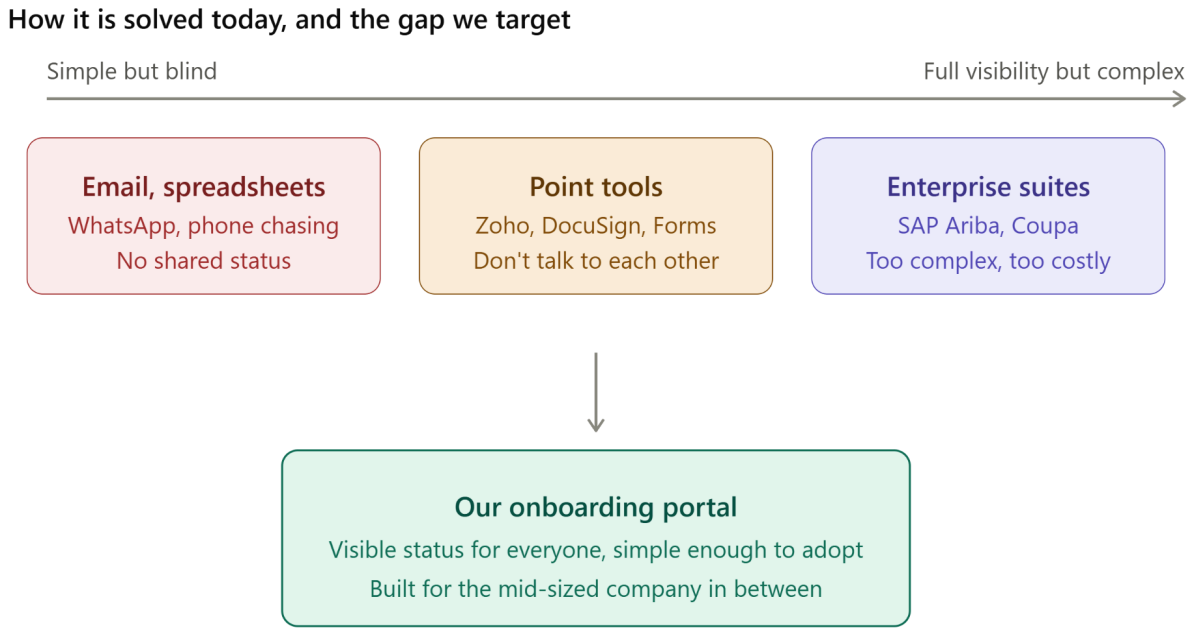
- ***Vendor self-registration*** (invite-only enforced); the full vendor lifecycle / RFP; payroll (stays in Workday); the vendor-abandons-onboarding flow; performance review and HR case management.
- ***Native mobile app*** (mobile web only); multi-language (English only in phase 1); IT/support ticket creation; new-department setup; admin adding departments.
- ***Auto-reminders and auto-escalation*** on SLA breach; OCR/format conversion (only the configured error message is shown); per-document sub-user assignment (one vendor login handles the whole submission).

Explicitly out of scope forever (for this product): end-to-end procurement suite functionality that would replace the buyer's ERP or payments system. We integrate with those, we don't rebuild them.

3.3 User Flows

This section maps the paths a user actually takes, from the moment a vendor is invited to the moment they go live. We cover three: the first-time onboarding path, the day-to-day department review, and what happens when something needs fixing.

The main flow reads top to bottom. The vendor submits everything once, the system validates and routes it, and all four departments review at the same time rather than one after another. Only once every department has approved does the request reach the Admin, whose final approval is an optional gate.



Reviewing in parallel instead of in sequence is what brings onboarding down from weeks to days. And because the whole thing runs on one shared record, every team and the vendor see the same live status the entire way through.

Flow 1: First-time vendor onboarding. This is the core new-user path.

Trigger: Procurement invites a vendor.

Steps: The vendor gets the invite link, signs up, and verifies through an OTP. They fill in the onboarding form and upload all the required documents in one go, then submit. The system checks for duplicates and validates the files, then routes the request to HR, Legal, Finance, and Procurement at the same time. Each department reviews its part and approves. Once all four are done, the Admin gives final approval, if that gate is switched on.

Success ending: **The vendor is onboarded and shows as Active.**

Where it can go wrong: A department rejects or flags the vendor, which sends it into resubmission or an Admin review. A required document is missing, so submission is blocked with a clear message. Or the vendor already exists (caught on the GSTIN), which is blocked with a duplicate error.

Flow 2: Department review. This is the flow each reviewer runs day to day.

Trigger: A submitted vendor lands in a department's queue.

Steps: The department user logs in and sees their pending tasks. They open the request in read-only mode and view or download the documents. They either approve it, reject it with a mandatory comment, or flag it to the Admin. Whatever they do, the status bar updates for everyone straight away.

Success ending: **That department's part of the review is cleared.**

Where it can go wrong: A user tries to act on something outside their own department and gets blocked (sent to an unauthorized screen). Or the SLA is running out, in which case the due-date indicator shows and the Admin is notified.

Flow 3: Resubmission (the recovery path). This is what happens when a reviewer needs a change.

Trigger: **A department requests changes.**

Steps: The SLA clock pauses. The vendor is notified and can see exactly which document or field needs fixing. They make the fix and resubmit with a short clarification. The clock resumes, with the due date pushed out by however long it was paused. The department then reviews the change.

Success ending: **The fix is accepted and the request carries on through the flow.**

Where it can go wrong: Repeated back-and-forth, which we track in analytics as the vendor's resubmission time, so a vendor who keeps missing the mark shows up in the data.

3.4 Screens and States

Every screen and the states it can be in. Real products spend most of their time in the non-happy states.

Screen	Purpose	Key states
Login / OTP verify	Authenticate all roles; vendor sets password via OTP	Empty, loading, error (bad OTP/expired), success
Vendor onboarding form	Collect details + documents	Draft (auto-saved), validation error (format/size/missing), submitted, read-only after submit
Vendor status view	Real-time department- and document-wise status	In review, changes requested, approved, halted, empty (nothing submitted yet)
Department review queue	List of pending vendor tasks	Empty (no tasks), loading, populated, SLA-due / SLA-breached indicators
Department review detail	Read-only vendor data + approve/reject/flag	Pending, approved, rejected (comment required), flagged, unauthorized if wrong role
Admin configuration	SLAs, triggers, managers, AI toggle, final-approval toggle	Default, edited/saved, validation error
Admin status + analytics dashboard	Oversight and metrics	Populated, partial ("—" when insufficient data), empty

Unauthorized	Block out-of-role access	Shown for any route outside the user's role/department

3.5 User Stories With Acceptance Criteria

Numbered; the full set expands from the scope list. A representative sample at the required level of detail:

Story 1. As a vendor, I want to submit my details and documents in one place, so that I don't have to email different teams separately.

- Given I am invited and verified, when I complete all mandatory fields and upload valid documents and click Submit, then my application locks to read-only, routes to all four departments in parallel, and my status view shows "In review" per department.
- Given a document is the wrong format or size, when I try to submit, then submission is blocked and the configured error message names the exact file and rule.
- Given a vendor with my GSTIN already exists, when I enter it, then submission is blocked with a clear duplicate-vendor error.

Story 2. As a department reviewer, I want to approve or reject only my department's documents, so that review happens in parallel and I can't act outside my remit.

- Given I am a Finance user, when I open a vendor request, then I see the vendor data read-only and can act only on Finance documents (PAN, GST, Bank Statement, Turnover).
- Given I reject, when I confirm, then a comment is mandatory, the vendor is notified, and the SLA clock pauses.
- Given I try to open the Admin area or another department's review, when I navigate there, then I am sent to /unauthorized.

Story 3. As an Admin, I want to give final approval after all departments clear, so that there is one controlled gate before a vendor goes live.

- Given all four departments have approved and the final-approval toggle is on, when I approve, then the vendor becomes Active and everyone's status bar shows Onboarded.
- Given the toggle is off, when the last department approves, then the vendor auto-onboards without my step.
- Given any department has flagged the vendor, when I review, then I can halt onboarding, which blocks all department actions until resolved.

3.6 Edge Cases

Situation	Expected behaviour
A Finance user tries to act in the Legal flow	Blocked at route and data level; redirected to /unauthorized
Vendor submits with a mandatory field or document empty	Submission blocked; inline validation names the missing item
SLA clock during a resubmission request	Clock pauses on change-request, resumes on resubmission; paused time excluded from breach counts

Vendor already exists in the buyer's database	Blocked on invite (email/name) and on GSTIN entry, with a clear error
Buyer tries to edit vendor-entered data	No write path exists for buyers; vendor data is read-only, enforced server-side
Vendor tries to view another vendor's record	No vendor-facing route accepts another vendor's id; data is isolated
Any list/detail screen has no data yet	A specific empty-state message is shown, never a blank screen
Admin halts a vendor mid-flow	All department actions on that vendor are blocked until the halt is lifted

3.7 Non Functional Requirements

- **Auth:** invite + OTP for vendors; role-based login for internal users. Roles: Admin, HR, Legal, Finance, Procurement, Vendor.
- **Platform:** responsive web (desktop + mobile web). No native app in v1.
- **Integrations:** none hard-wired in the prototype; file storage and email are mocked (in-app notifications, OTP shown on screen). Real build would integrate email + document storage.
- **Constraints:** must be simple enough for non-technical vendors and reviewers to use without training, this is the core adoption requirement.
- **Accessibility:** standard web accessibility (labelled fields, keyboard navigation, clear error text) as the minimum bar.

PART 4: OPEN QUESTIONS, ASSUMPTIONS AND RISKS

Open questions

Question	Why it matters	How we'll resolve it
Should Admin final approval be mandatory or optional?	Changes whether vendors auto-onboard after dept approval	Built as a toggle; validate the default with users
How to handle SLA when both dept managers are on leave?	Could stall a request with no owner	Out of scope for v1; flagged as a known gap

Assumptions

Assumption	What breaks if it's wrong
Access is provisioned within 3 working days (process SLA, not system-enforced)	If slower, onboarding cycle-time target is missed for reasons outside the tool
Vendors can submit all mandatory documents in one go	If they can't, the single-submission model needs a partial-save workflow
The four departments (HR, Legal, Finance, Procurement) cover all required reviews	A missing review type (e.g. Compliance) would need a new department
Vendor base is willing to use a portal at all	Our interviews flagged some vendors as non-tech-savvy; low adoption would undercut the model

Risks

- Adoption risk: if non-technical vendors won't use the portal, onboarding falls back to email. Mitigation: keep the vendor form minimal and allow Procurement to fill it on the vendor's behalf.
- Scope-creep risk: pressure to add lifecycle features would recreate the unusable enterprise suite. Mitigation: the explicit out-of-scope list above.
- Compliance risk: mocked document handling in the prototype means no real validation of document authenticity. Mitigation: clearly labelled as prototype-level; real build adds verification.